Sammy Al Hashemi

Recent engineering graduate looking for entry-level positions in software development/machine learning.

Paris, France

Career History

2018 **Microsemi Corp. – Device Characterization Engineering Intern**

July 2017 to August 2018 (1.1 years)

- July 2017 - August 2018
- Education Characterized the Mathblock component of Microsemi's G5 FPGA product as well as performance-related AC tests on other combinatorial and sequential elements found in the FPGAs that went to market.
  - Example components include look-up tables, carry chains, digital I/O's.
  - Tested chains on various corners of the FPGA fabric.
- Inside Libero, I created Verilog test structures, making use of JTAG to read input from test programs created using a company code.
- Wrote sections of the characterization report found in the official product manual used to facilitate customer knowledge in our device.

Education

2019 **Bachelor of Applied Science and Engineering in physics from University of Toronto**

September 2014 to April 2019 (4.6 years)

Thesis

Investigation into the Applicability of Grover's Algorithm
Grover's Algorithm is a quantum algorithm that can, under the right conditions, search an unstructured database faster than any classical algorithm. In fact, it achieves an exponential speedup. The catch is the 'right conditions' description. Through vigorous literature review and implementations of small implementations of Grover's, the applicability of the algorithm was investigated into its runtime, the real-world hardware requirements, and if it could be implemented without an explicit oracle circuit.

Projects

Data Charter GUI in Python (https://github.com/Sammyalhashe/Charter-cp)

- Built a charting application that reads data interfaced by a *ni daqmx* (National Instruments) board.
- The project was treated as a fullstack application with the frontend being the GUI while the backend was the code that read data that was fed to the frontend.
- The GUI was built using *pyqt*, a `python` abstraction of the popular *QTc* library for building quality cross-platform GUI's.
- The backend implemented National Instruments *nidaqmx* `package` that allowed easy reading from various channels of the board.
- The application could read multiple channels at once and either:
  - Plot them vs. time simultaneously.
  - Plot them vs each other (ie. channel 1/3/4 vs. channel 2)
  - Save the read data by either exporting images or by exporting a csv file.
- This application was built to replace the previously used Labview implementation of the charter in University of Toronto's Advanced Physics Labs, due to Labview's lack of easy flexibility.
- This application is currently in production at the University of Toronto.

Personal Website (https://github.com/Sammyalhashe/Website2)

- Single page application written with `Angular6`.
- Uses Firebase's Cloud Storage to store and asynchronously fetch larger files such as images.
- Uses complex `angular` routing features such as lazy-loading.
- RxJS usage for asynchronous data handling (also migrated from RxJS5 to RxJS6)

Dashcam Speed Visualizer (https://github.com/Sammyalhashe/commai-source)

- Comma.ai's dashcam product has the capability to sample data every second, of which includes speed, coordinates, and distance travelled since the trip began.
- Using 527 trips, stored as *JSON*, I created a performant webapp that simultaneously displayed all 527 and their speed distribution on an interactive map. See https://dashcamroutes.herokuapp.com/.
- I stored trips in a Mongo Cloud Atlas database in three collections, all dependent on the zoom of the map.
  - `Python` was used to preprocess the data, initially separating them by number of datapoints into separate directories then uploaded to the database using a `shell script`.
  - Client-side caching was implemented so that the data did not need to be downloaded multiple times.
- An `*Angular2+*` frontend was implemented with an *ExpressJS* backend with *Mongoose* acting as the interface between Mongo and the app.

Inferred Skills

After reading this CV, HireFirst believes Sammy has experience with AngularJS, C, Python, Shell Scripting and Tcl.